Data-Driven Urban Metro Train Frequency and Capacity Optimization Using Transport for London NUMBAT Data

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***Abstract*—Urban metro demand varies strongly across the day and across the network, while train frequencies are fixed by timetables. This paper presents a data-driven decision-support system that uses Transport for London's NUMBAT open data to (i) profile and validate link-level passenger loads and scheduled train frequencies, and (ii) solve three Operations Research models: an integer program for the minimum train frequency that carries observed demand, a two-stage mixed-integer program that allocates a limited link-service budget to maximise demand served, and a weighted goal program that balances operating effort, demand satisfaction, adherence to the existing service pattern and a target capacity utilisation. The NUMBAT 2025 workbook yields 116,544 link–interval observations (1,214 links, 19 lines, 96 quarter-hours). A consistency check between the two link sheets finds 5,770 observations (0.41% of demand) where passengers are reported on a link with zero scheduled trains; these are flagged and handled by an explicit, user-selectable policy. With published rolling-stock capacities and an 80% utilisation target, the minimum-frequency model needs 57.8% fewer link-train allocations than the timetable network-wide, showing that observed frequencies are set by headway standards rather than by volume alone. Scenario analysis quantifies the effect of 10–20% demand growth, reduced service budgets and alternative utilisation targets. The models are implemented in Pyomo with the HiGHS solver and delivered through an interactive Streamlit dashboard.**

***Index Terms*—train frequency setting, integer programming, goal programming, capacity allocation, public transport, NUMBAT, Transport for London, Pyomo, HiGHS, decision support**

I. Introduction

Metro systems carry large and highly variable passenger volumes. Demand peaks sharply in the morning and evening, differs between lines and between consecutive segments of the same line, and falls to very low levels late at night. Train frequency, on the other hand, is planned in advance and constrained by rolling stock, crews and signalling. When the two are misaligned, passengers experience overcrowding on busy segments while trains on quiet segments run with spare capacity [2], [3].

Why this matters. From a passenger perspective, matching capacity to demand reduces crowding and keeps service levels consistent. From an operator perspective, it identifies demand–service mismatches and avoids unnecessary service in low-demand periods. For decision makers, it replaces fixed rules of thumb with quantitative, reproducible recommendations whose sensitivity to demand growth and resource limits can be tested before any change is made.

Transport for London (TfL) publishes the NUMBAT dataset [1], which reports, for every link between consecutive stations and every 15-minute interval of a typical day, both the passenger load and the number of scheduled trains. Having demand and supply at the same resolution makes NUMBAT well suited to frequency and capacity optimisation.

In this work we first convert the wide-format NUMBAT workbook into a clean, validated table of link–interval observations, including a check for passengers reported on links with no scheduled train. On this data we formulate and solve three Operations Research models: an integer program for the minimum frequency, a mixed-integer program for allocating a limited service budget, and a weighted goal program that balances several planning goals. We then test how the solutions respond to demand growth, budget cuts and different utilisation targets, and deliver the whole workflow as an interactive dashboard that works directly on the workbook downloaded from TfL.

The remainder of the paper is organised as follows. Section II reviews related work, Section III describes the dataset and Section IV the research questions. Section V presents the methodology and models, Section VI the implementation and user interface and Section VII the results. Section VIII

discusses limitations and Section IX concludes.

II. Related Work

Frequency setting is a classical stage of transit planning, following network design and preceding timetabling, vehicle scheduling and crew scheduling [3]–[5]. The simplest and most widely used rule is the *maximum-load-point* method, which sets the frequency on a route to the peak load divided by the desired occupancy per vehicle [2]. Ceder [2] refines this with load-profile methods that use the load on every segment rather than only at the peak point, which is the link-level view adopted here. Furth and Wilson [6] formulated frequency allocation across routes as a resource-constrained optimisation that maximises net social benefit subject to a fleet budget, the conceptual basis of our Model 2.

When several conflicting objectives must be balanced, goal programming [7]–[9] is a natural tool: each objective becomes a goal with a target, and weighted deviations from the targets are minimised. Normalisation of the deviations is essential when goals are measured in different units [8], [9].

Open-source algebraic modelling languages such as Pyomo [11], [12] and high-performance open solvers such as HiGHS [13] make it practical to embed these models in interactive decision-support tools. Our work combines these elements with a publicly available, high-resolution demand dataset and places particular emphasis on data validation, which is often assumed rather than demonstrated in applied studies.

III. Dataset

A. Source and coverage

NUMBAT is published by TfL Public Transport Service Planning at https://crowding.data.tfl.gov.uk/ [1]. It is produced from smartcard ticketing data and gateline entry/exit counts: each journey's route is estimated and assigned to trains and stations. The data represent a *typical* autumn day (here the 2025 Tuesday–Thursday average, file `NBT25TWT_Outputs.xlsx`) from 05:00 to 04:59, assume a full timetabled service, and cover the London Underground, London Overground, Docklands Light Railway (DLR), Elizabeth line and London Trams.

B. Workbook structure

The workbook contains nine worksheets; this work uses the two link sheets in Table I. Each is in *wide* format: two title rows (hour and quarter-hour indices), a header row with the field names, then one row per link with a daily total, six time-of-day totals and 96 quarter-hour columns labelled `0500-0515` to `0445-0500`.

C. Variables used

A *link* is a directed segment between consecutive stations on one line (e.g. Victoria line southbound, Green Park → Victoria), identified by a unique code such as `GPKu_VIC_SB>VICu_VIC_SB@VIC`, with line, direction, sequence order, NLC/ASC station codes and station names.

TABLE I
NUMBAT worksheets used in this work

Sheet	Rows	Content and use
`Link_Loads`	1,214	Passengers on each link per 15 min: demand D_i
`Link_Frequencies`	1,214	Scheduled trains on each link per 15 min: observed supply f_i

For link ℓ and interval t, the *link load* is the number of passengers traversing the link during the interval, and the *link frequency* is the number of trains traversing it during the same interval. Frequencies are *trains per 15 minutes*, not trains per hour. Because a passenger is counted on every link they ride, summing link loads gives passenger-link trips, not unique journeys.

D. Summary statistics

After preprocessing (Section V-B) the dataset contains 1,214 links on 19 lines between 482 stations, giving 116,544 link–interval observations, of which 116,352 (99.8%) have validated demand and frequency. Across the network the day totals 39,411,202 passenger-link trips and 256,422 link-train allocations. The AM and PM peak periods together account for 52.0% of daily demand.

IV. Research Questions

- **Q1** **Demand:** what are the temporal and spatial patterns of passenger demand across the network?
- **Q2** **Service:** how does the existing train frequency correspond to passenger demand?
- **Q3** **Frequency optimisation:** what train frequency is required to satisfy demand under capacity constraints? (Model 1)
- **Q4** **Capacity allocation:** given limited service capacity, how should trains be allocated across segments and time periods? (Model 2)
- **Q5** **Multi-objective planning:** how can frequency be planned while balancing operating effort, crowding and service quality? (Model 3)
- **Q6** **Sensitivity:** how does the optimal solution change when demand, capacity targets or resources change?

V. Methodology

Fig. 1 summarises the pipeline. The uploaded workbook is the only source of operational observations; every assumption not contained in it (capacity per train, utilisation target, frequency bounds and budget) is an explicit user input.

A. Data parsing

NUMBAT downloads differ between years and day types, so nothing is read from fixed rows or columns. The parser (i) finds the link sheets by name or by their title cell; (ii) selects as header row the row that matches the most known field names (with aliases such as *Dir*/*Direction*); and (iii) recognises interval columns by the pattern `HHMM-HHMM`, accepting only 15-minute spans. Intervals are ordered from 05:00 across

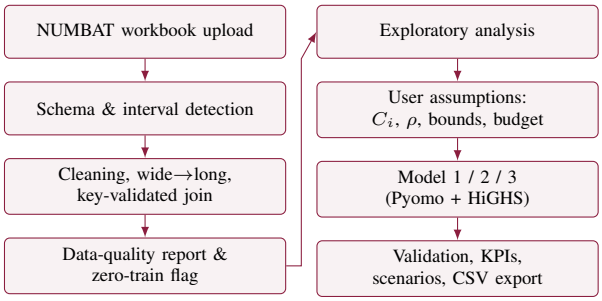


Fig. 1. Processing and optimisation pipeline implemented in the dashboard.

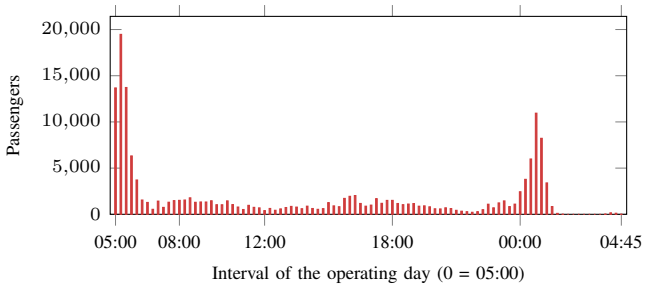


Fig. 2. Passengers reported on link–intervals that have zero scheduled trains, by interval (all lines).

midnight to 04:59, the NUMBAT operating day. Ambiguous header matches are reported rather than resolved silently.

B. Preprocessing and validation

The two link sheets are cleaned and reshaped from wide to long format (one row per link and interval), then joined. The rules are:

- **Join key.** The first candidate key that exists and is unique in both sheets is used: the link code, then line + direction + NLC codes, then line + direction + station names. The join is on key and interval, never on row position. For the 2025 workbook the link code is unique in both sheets.
- **Missing is not zero.** Empty, non-numeric and negative cells are flagged and excluded; they are never replaced by zero. Genuine zeros are kept.
- **Duplicates.** Exact duplicate rows are dropped once; records that share a key with different values are *ambiguous* and excluded, never averaged.
- **Cross-sheet consistency.** For every matched link the descriptive fields of the two sheets are compared. 19 station labels differ only by a branch qualifier (e.g. "Kennington (Bank)" vs. "Kennington") and 13 links differ only in sequence number; these are reported and kept. 2 links have genuinely conflicting identities (one link's destination is "Liverpool Street" in one sheet and "Hackney Downs" in the other; another has a different NLC) and are excluded (192 observations).
- **Totals.** The daily *Total* column is checked against the sum of the 96 interval values (consistent for every link) but is never used as interval demand.

C. Demand reported with zero scheduled trains

A physically consistent observation cannot have passengers on a link in an interval with no train. The dataset contains 5,770 such link–intervals (4.96% of matched observations), carrying 163,247 passengers (0.41% of demand). Fig. 2 shows that they concentrate at the start and end of the traffic day: 33% of this demand falls between 05:00 and 06:00 and 22% after midnight. For 85% of the demand, the same link has trains in the adjacent interval. This indicates a period-boundary effect: NUMBAT's assigned loads and the timetable assign a train run to slightly different quarter-hours, and the averaging over several days spreads small loads into neighbouring periods.

TABLE II
DEFAULT CAPACITY PER TRAIN (EDITABLE IN THE DASHBOARD)

Line	Stock (cars)	C	Source
Bakerloo	1972 (7)	851	[14]
Central	1992 (8)	1,047	[14]
District, H&C, Circle	S7 (7)	1,045	[14]
Jubilee	1996 (7)	964	[14]
Metropolitan	S8 (8)	1,176	[14]
Northern	1995 (6)	752	[14]
Piccadilly	1973 (6)	798	[14]
Victoria	2009 (8)	986	[14]
Waterloo & City	1992 (4)	506	[14]
Elizabeth line	Class 345 (9)	1,497	[15]
LO Mildmay, Windrush	Class 378 (5)	840	[16]
LO Liberty, Weaver	Class 710 (4)	678	[17]
LO Suffragette, Lioness	Class 710 (4)	678	[17]
DLR	B07/B2K (3)	688	[18]
London Trams	Variobahn	206	[19]

Every such observation is flagged (`zero_train_demand`) and reported by line and interval in the Data Quality tab. Because the right treatment is a modelling decision, it is exposed as a policy:

- **Exclude (default):** the observations are removed from optimisation and the removed demand is disclosed with every result;
- **Include:** the demand is kept, so models may add service where the timetable has none.

Section VII compares both policies.

D. User-supplied parameters

Capacity per train C_i. NUMBAT does not contain vehicle capacity. Default values per line were taken from TfL rolling-stock data sheets [14]–[16] and, where no TfL sheet was found, from secondary sources [17]–[19] (Table II). For the Underground they are seats plus standing room at 5 passengers/m^2, which TfL calls the "maximum observed standing capacity". All values can be edited in the dashboard, or replaced by one uniform value, which is then disclosed with every result.

Target utilisation ρ. The usable capacity is $C_i^{\text{eff}} = \rho C_i$ with $0 < \rho \leq 1$, a planning buffer below rated capacity. The default is $\rho = 80\%$, a modelling choice and not a TfL value.

Frequency bounds. Optional minimum l and maximum u trains per link per 15 minutes, with $l \leq u$ enforced. The

minimum can be applied only where the link is observed in service, so that overnight closures are not forced open.

Link-service allocation budget. The workbook gives trains per link, not the number of physical trains, and one train traverses many consecutive links. We therefore define a *link-service allocation budget*

$$F_t = \left\lfloor \beta \sum_{i \in t} f_i \right\rfloor, \tag{1}$$

where β is a user-chosen fraction of the observed allocations in interval t. F_t is a modelling resource, not a fleet count.

E. Notation

Index i denotes one observation (a link in a 15-minute interval) and t an interval; $i \in t$ means that observation i lies in interval t. The data are the demand D_i and observed trains f_i; the parameters are the capacity per train C_i, the target utilisation ρ (so $C_i^{\text{eff}} = \rho C_i$), the bounds l_i, u_i and the budget F_t. The decision variables are the number of trains x_i (integer) and the passengers served y_i.

F. Model 1: minimum train frequency (integer programming)

$$\min \quad \sum_i x_i \tag{2}$$

$$\text{s.t.} \quad C_i^{\text{eff}}\, x_i \geq D_i \qquad \forall i \tag{3}$$

$$\sum_{i \in t} x_i \leq F_t \qquad \forall t \tag{4}$$

$$l_i \leq x_i \leq u_i, \quad x_i \in \mathbb{Z}_{\geq 0} \qquad \forall i \tag{5}$$

The budget constraint (4) is optional. Without it, each observation can be solved separately and the optimum is $x_i = \max\left(l_i, \lceil D_i / C_i^{\text{eff}} \rceil\right)$; the solver output is tested against this. If a maximum frequency or the budget cannot cover demand, the model is reported as infeasible and the conflicting links or intervals are listed. Constraints are never relaxed silently.

G. Model 2: capacity-constrained service allocation (MILP)

$$\max \quad \sum_i y_i \tag{6}$$

$$\text{s.t.} \quad y_i \leq D_i \qquad \forall i \tag{7}$$

$$y_i \leq C_i^{\text{eff}}\, x_i \qquad \forall i \tag{8}$$

$$\sum_{i \in t} x_i \leq F_t \qquad \forall t \tag{9}$$

$$l_i \leq x_i \leq u_i, \quad x_i \in \mathbb{Z}_{\geq 0}, \quad y_i \geq 0 \qquad \forall i \tag{10}$$

The objective does not penalise trains, so an optimum may contain trains that serve nobody. A second stage [10] therefore minimises $\sum_i x_i$ while keeping $\sum_i y_i$ at its optimal value Y^*. An interval is reported as *binding* when $\sum_{i \in t} x_i = F_t$ and demand in that interval is left unmet, i.e. the budget actually limits service.

H. Model 3: weighted goal programming

Model 3 keeps the hard constraints $y_i \leq D_i$, $y_i \leq C_i x_i$ and $l_i \leq x_i \leq u_i$ (rated capacity is the physical limit) and expresses four planning goals in the standard goal-programming form, with non-negative deviation variables n (under the target) and p (over the target):

$$\text{G1 effort:} \quad \sum_i x_i + n_1 - p_1 = E \tag{11}$$

$$\text{G2 demand:} \quad \sum_i y_i + n_2 = \sum_i D_i \tag{12}$$

$$\text{G3 pattern:} \quad x_i + n_{3i} - p_{3i} = f_i \qquad \forall i \tag{13}$$

$$\text{G4 utilisation:} \quad \rho C_i x_i + n_{4i} - p_{4i} = D_i \qquad \forall i \tag{14}$$

Here $E = \sum_i \max\left(l_i, \lceil D_i / C_i^{\text{eff}} \rceil\right)$ is the fewest trains that carry all demand at the target utilisation, and the observed frequency f_i is the target service pattern. Each unwanted deviation is divided by its target (percentage normalisation [8], [9]), giving dimensionless goal values

$$z_1 = \frac{p_1}{E}, \qquad z_2 = \frac{n_2}{\sum_i D_i},$$
$$z_3 = \frac{\sum_i (n_{3i} + p_{3i})}{\sum_i f_i}, \qquad z_4 = \frac{\sum_i (n_{4i} + p_{4i})}{\sum_i D_i}, \tag{15}$$

and the model minimises their weighted sum

$$\min \quad \sum_{k=1}^{4} w_k z_k. \tag{16}$$

The weights w_k start equal and are user settings. In G4, n_{4i} is load above the target ρC_i and p_{4i} is unused target capacity; unmet demand (G2) arises only above the rated capacity C_i. A very small equal weight on every goal keeps solutions Pareto-efficient when a user sets a weight to zero.

I. Scenario and sensitivity analysis

Each model can be re-solved under: (A) baseline; (B) $D_i' = 1.10 D_i$; (C) $D_i' = 1.20 D_i$; (D) the allocation budget reduced by a chosen percentage; and (E) alternative utilisation targets ρ. Scenario D reduces the modelled budget, not TfL's physical fleet. Infeasible scenarios are reported as infeasible.

J. Solution verification

HiGHS [13] is called through Pyomo [12] with a time limit and a relative MIP gap of 10^{-4}. A result is shown only if a solution exists. It is labelled "optimal" only if HiGHS proves optimality and the primal and dual bounds agree. Independent checks then verify integrality, bounds, $y_i \leq D_i$, $y_i \leq$ capacity $\times\, x_i$, demand coverage (Model 1) and the budget in every interval. A failed check suppresses the result.

VI. Implementation and User Interface

A. Architecture

The system is written in Python with a deliberately small codebase (Table III): pandas and NumPy for data processing [20], Pyomo with HiGHS for optimisation, Plotly for charts, NetworkX [21] for the network view, Streamlit [22] for the interface and pytest for 24 automated tests.

TABLE III
CODE MODULES

Module	Responsibility
`data_loader.py`	Sheet, header-row, field-alias and interval detection; cover notes
`preprocessing.py`	Cleaning, reshaping, key-validated join, quality metrics, zero-train flag, readiness
`optimization.py`	Models 1–3, result tables, KPIs, post-solve validation
`analysis.py`	Descriptive summaries, interpretation text, scenario runner
`visualization.py`	Plotly charts and schematic network
`app.py`	Streamlit dashboard (five tabs)
`tests/`	24 unit and integration tests

```
merged["zero_train_demand"] = (merged["eligible_service"]
    & (merged["frequency"] == 0) & (merged["demand"] > 0))
grp = merged.groupby("link_key")
adjacent = pd.Series(False, index=merged.index)
for step in (1, -1): # trains on the same link in the next/
     previous interval?
    nxt = grp["interval_order"].shift(-step) == merged["
         interval_order"] + step
    adjacent |= nxt & (grp["frequency"].shift(-step) > 0)
merged["zero_train_adjacent_service"] = merged["
     zero_train_demand"] & adjacent
```

Listing 1. Flagging zero-train demand (`preprocessing.py`).

B. Code excerpts

Listing 1 shows how the zero-train observations are flagged, and Listing 2 the core of Model 2, including the second stage.

C. Dashboard

A user uploads the unmodified workbook. The sidebar holds the scope (lines, directions, time window), capacity assumptions, service constraints and the zero-train policy. Five tabs follow the analysis workflow: **Overview** (readiness per model, scope KPIs, demand and service profile, cover-sheet notes; Fig. 3), **Data Quality** (profiling, preprocessing log, exclusions, zero-train analysis and the processed-data download; Fig. 4), **Explore Data** (top links and lines, peak vs. off-peak, demand heatmap, demand vs. service, mismatch tables and a schematic network; Fig. 5), **Optimization** (model choice, formulation, parameters, the *Optimize* button, KPIs, generated interpretation, before/after charts, goal deviations, validation checks and CSV download; Fig. 6) and **Sensitivity Analysis** (scenario selection, comparison table, chart and CSV; Fig. 7). Every result is displayed together with the dataset, scope and assumptions that produced it. Results are invalidated when a different workbook is uploaded.

VII. RESULTS

Unless stated otherwise, results use the default capacities (Table II), $\rho = 80\%$, no minimum or maximum frequency, a budget equal to the observed allocations ($\beta = 1$) and the *exclude* zero-train policy. All numbers are produced by `report/generate_report_data.py` from the same code as the dashboard.

```
tech_ub = np.ceil(D / Ce - 1e-9) # more trains cannot serve
     more
m = _base_model(w, _x_bounds(w, tech_ub), budgets) # x_i,
    budget rows
m.y = pyo.Var(m.I, within=pyo.NonNegativeReals,
          bounds=lambda _, i: (0, D[i]))
m.served_cap = pyo.Constraint(m.I,
          rule=lambda m, i: m.y[i] <= Ce[i] * m.x[i])
m.obj = pyo.Objective(expr=sum(m.y[i] for i in m.I), sense=
    pyo.maximize)
sol1 = _solve(m, s) # stage 1: max served demand
best = sum(_values(m.y, len(w)))
m.obj.deactivate() # stage 2: same service, fewer trains
m.keep_served = pyo.Constraint(
   expr=sum(m.y[i] for i in m.I) >= best - max(1e-8 * best,
        1e-4))
m.obj2 = pyo.Objective(expr=sum(m.x[i] for i in m.I), sense
    =pyo.minimize)
sol2 = _solve(m, s)
```

Listing 2. Two-stage Model 2 in Pyomo (`optimization.py`).

TABLE IV
DAILY DEMAND (THOUSAND PASSENGER-LINK TRIPS), OBSERVED LINK-TRAIN ALLOCATIONS, PASSENGERS PER ALLOCATION AND OBSERVED LOAD FACTOR $D_i/(C_i f_i)$ IN % OF RATED CAPACITY, BY LINE

Line	Links	Demand	Alloc.	Pass./alloc.	Mean LF	Max LF
Northern	114	5,493	38,919	141	17.4	90
District	118	4,942	29,530	167	14.1	71
Elizabeth Line	84	4,892	13,124	373	20.8	119
Central	98	4,266	26,945	158	11.8	76
Jubilee	52	3,503	20,395	172	16.0	65
Piccadilly	104	3,342	28,435	118	12.6	57
Victoria	30	2,976	16,228	183	17.2	66
H&C and Circle	94	2,408	14,966	161	15.9	83
DLR	99	1,523	18,989	80	11.8	68
Metropolitan	72	1,470	12,558	117	8.6	66
Bakerloo	48	1,317	11,522	114	11.3	44
LO Windrush	56	1,106	9,070	122	14.2	94
LO Mildmay	54	958	7,040	136	16.1	96
LO Weaver	52	690	3,612	191	27.0	156
LO Lioness	36	270	2,664	101	14.5	57
LO Suffragette	24	201	1,692	119	17.2	82
Waterloo & City	2	50	605	83	14.9	70
LO Liberty	4	3	128	25	2.2	8
London Trams	71	0	0	–	–	–

A. Q1: demand patterns

Fig. 8 shows the network demand and service profiles. Demand peaks at 08:15–08:30 (1,050,908 passenger-link trips in one quarter-hour), with a second, broader evening peak around 17:45 (1,050,013). Table IV breaks demand down by line. The busiest single link is Farringdon → Liverpool Street LU (Elizabeth Line) with 193,615 passengers per day (Table V).

B. Q2: how service corresponds to demand

Service follows the shape of demand (Fig. 8) but is much flatter: trains run at near-peak frequencies through the inter-peak period, when demand is far lower. Comparing demand with the observed effective capacity $f_i\,\rho\,C_i$ (only where $f_i > 0$; zero-train records are treated in Section V-C), 226 link–intervals on 64 links of 7 lines carry more than 80% of rated capacity. They are concentrated on the LO Weaver (163) and Elizabeth Line (41) lines. The highest load factor, 156%,

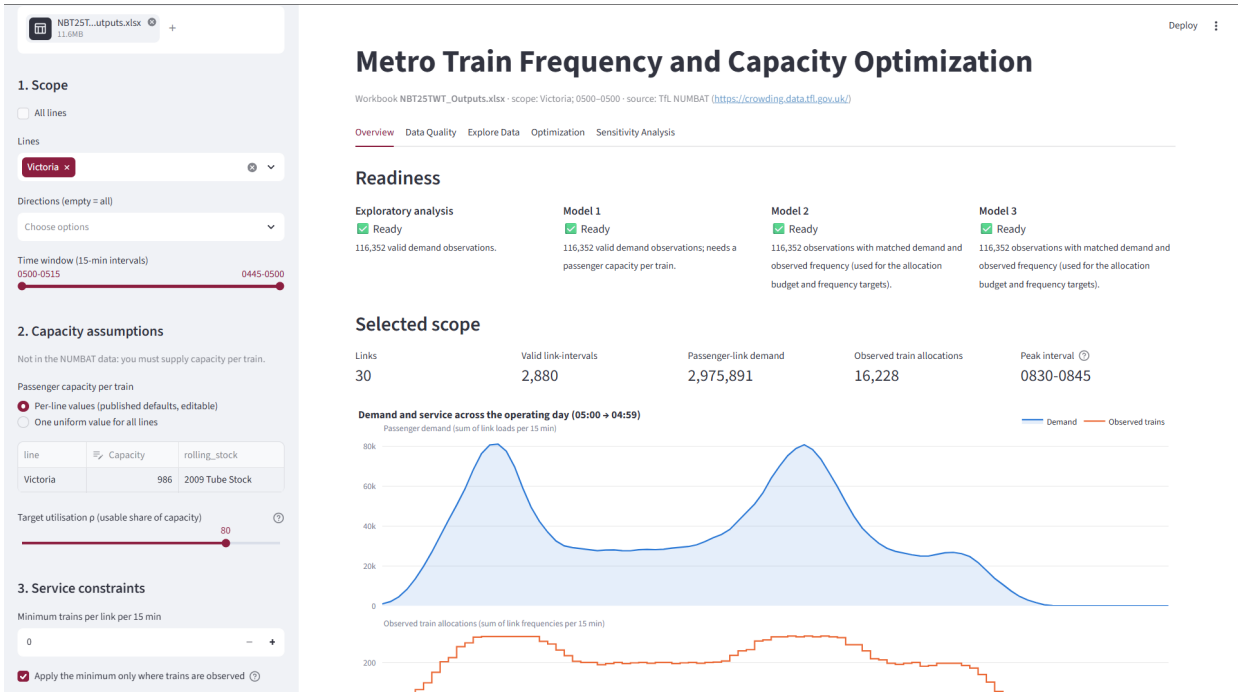


Fig. 3. Overview tab: sidebar inputs, readiness of each model, scope KPIs and the demand/service profile.

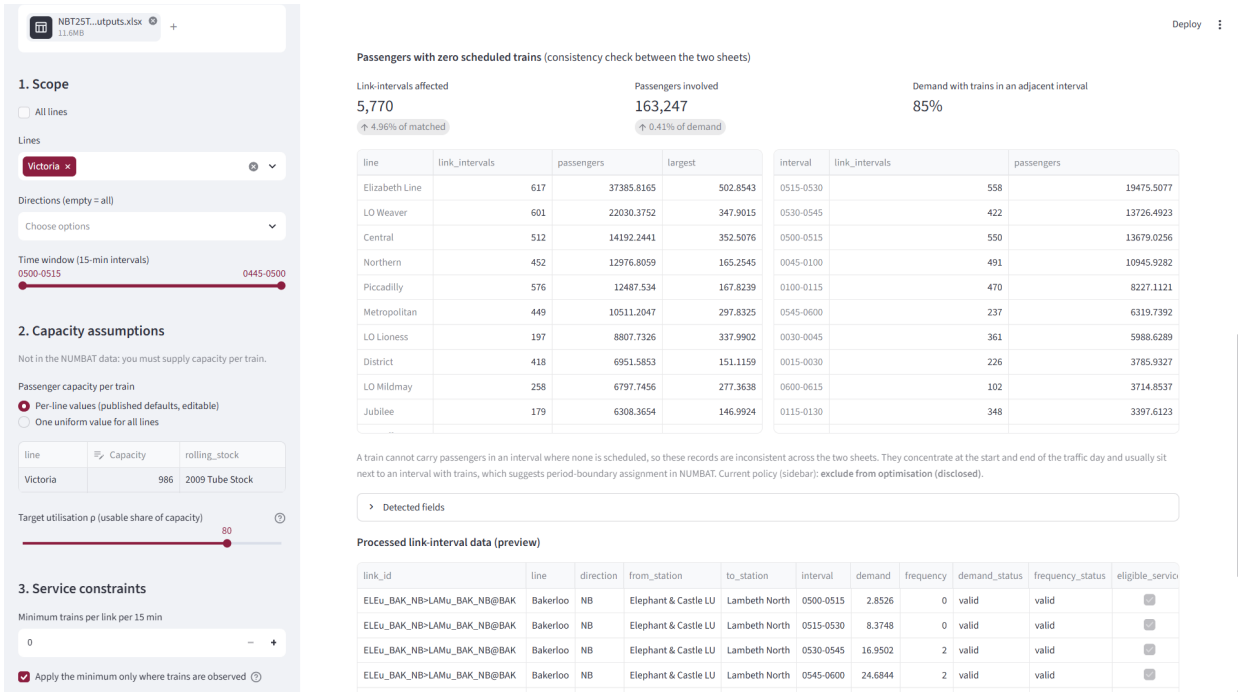


Fig. 4. Data Quality tab: sheet profile, join coverage, preprocessing log and the zero-train analysis.

occurs on the LO Weaver line. That value is partly an artefact of the conservative 4-car capacity assumed in Table II, since some peak services there run as 8-car pairs. The largest excess in passengers is on Stratford → Whitechapel (Elizabeth Line, 08:30) at 119% of the Class 345 full-load capacity. At the other extreme, 51,901 link–intervals with trains (61.6%) run below 15% of rated capacity. Mean load factors (Table IV) are low because they average over the whole day, while the maxima show where peak crowding occurs. The London Trams rows of this workbook contain zero demand and zero frequency in every interval, so they are excluded from the line comparisons.



Fig. 5. Explore Data tab: demand heatmap and schematic network topology (not a geographic map).

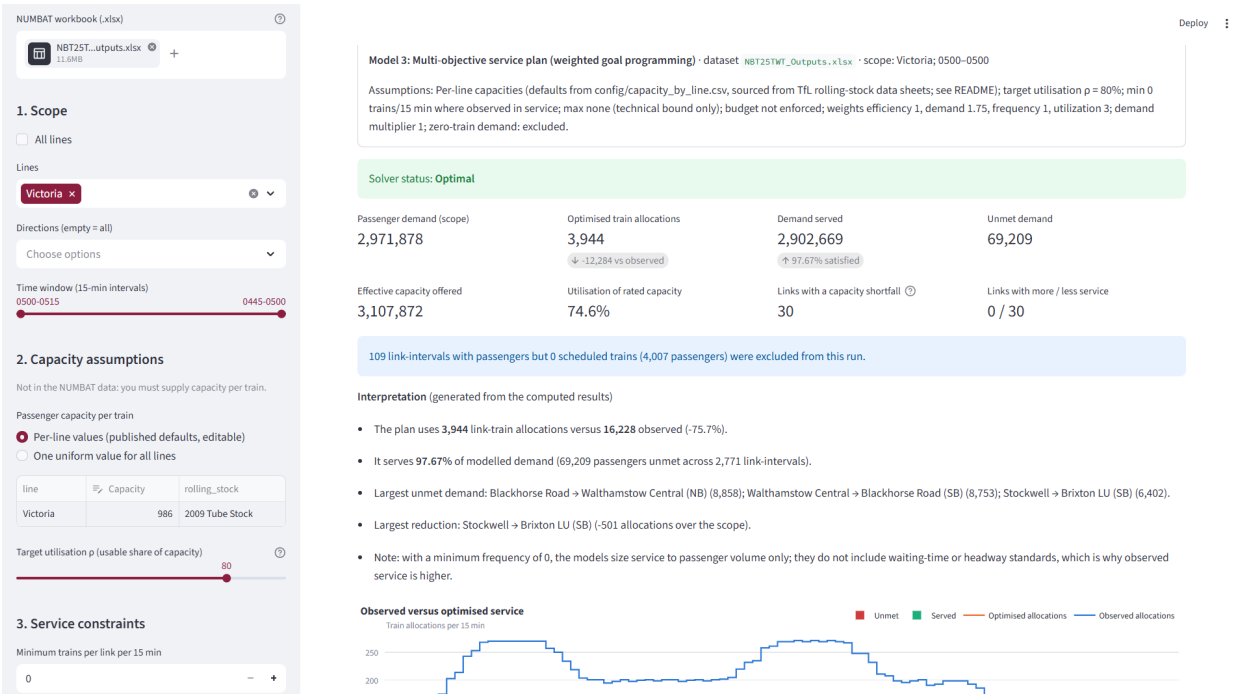


Fig. 6. Optimization tab: assumptions, solver status, KPIs, interpretation and before/after comparison.

C. Q3: minimum frequency (Model 1)

Network-wide, Model 1 solves all 110,582 eligible observations in 8.4 s and needs 108,124 link-train allocations against 256,422 observed, i.e. 57.8% fewer. Table VI shows the ratio per line. The ratio is highest on the LO Liberty line (100%) and lowest on the Victoria line (31%). The reduction does not mean that TfL over-supplies service. Model 1 sizes service to passenger volume only, while real timetables also provide short waiting times and regular headways. This is shown by the minimum-frequency experiments in the case study below.

Effect of the zero-train policy. With the *include* policy, Model 1 needs 113,885 allocations, 5,761 more than under

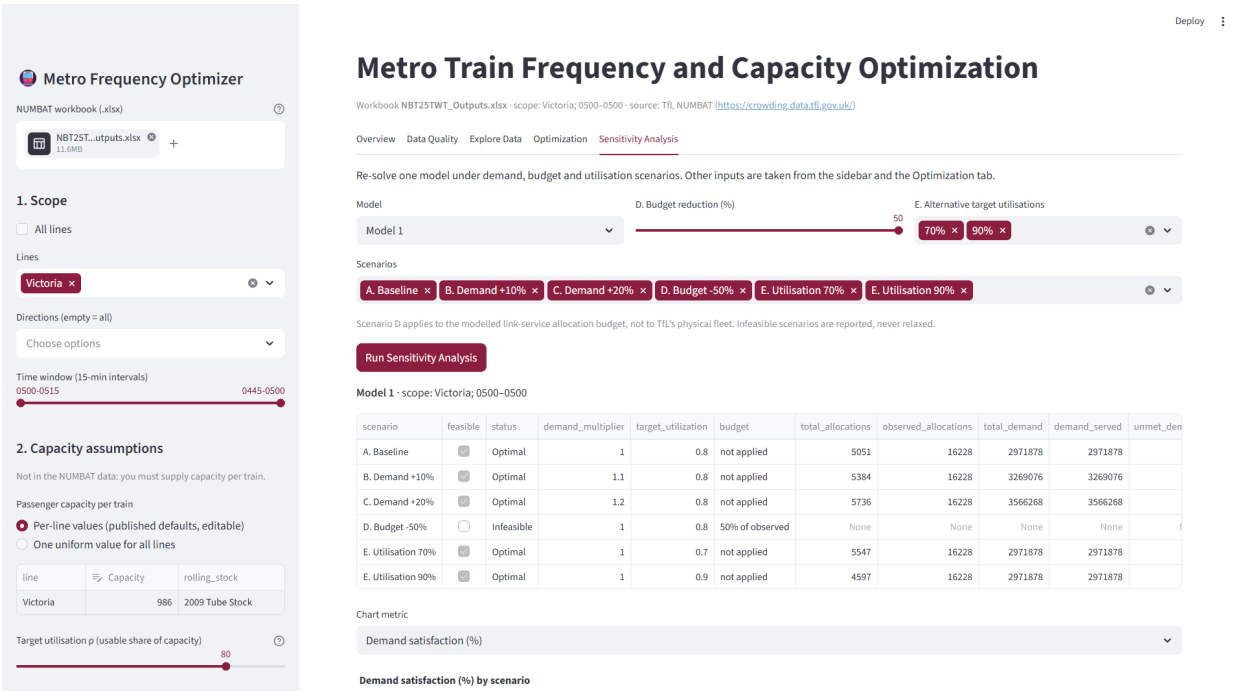


Fig. 7. Sensitivity Analysis tab: scenario comparison table and chart.

TABLE V
TEN HIGHEST-DEMAND LINKS (PASSENGERS PER DAY AND IN THE BUSIEST QUARTER-HOUR)

#	Link	Line	Daily	Peak 15 min
1	Farringdon → Liverpool Street LU	Elizabeth Line	193,615	6,455
2	Liverpool Street LU → Farringdon	Elizabeth Line	190,408	6,299
3	Whitechapel → Liverpool Street LU	Elizabeth Line	189,671	7,345
4	Liverpool Street LU → Whitechapel	Elizabeth Line	187,767	7,229
5	Tottenham Court Road → Farringdon	Elizabeth Line	184,833	5,529
6	Farringdon → Tottenham Court Road	Elizabeth Line	184,010	5,426
7	Oxford Circus → Warren Street	Victoria	160,324	5,447
8	Victoria LU → Green Park	Victoria	159,607	5,151
9	Green Park → Oxford Circus	Victoria	159,562	4,574
10	Tottenham Court Road → Bond Street	Elizabeth Line	159,290	4,991

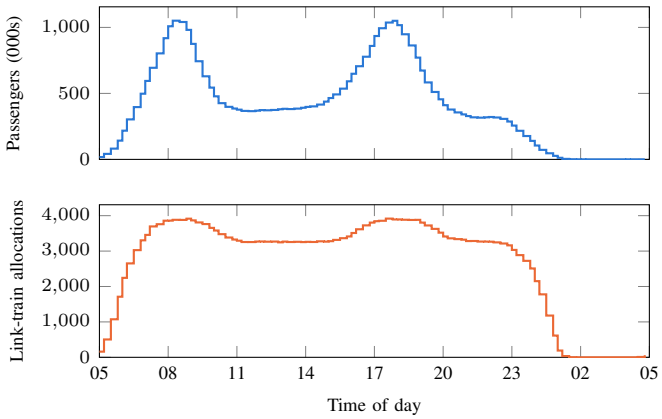


Fig. 8. Network-wide link demand (top) and scheduled link-train allocations (bottom) per 15 minutes, 05:00–04:59.

exclude. These extra trains would run in quarter-hours where the timetable has none, to carry only 0.41% of demand. Treating the inconsistent records uncritically would therefore distort the recommendations.

D. Case study: the Victoria line

The Victoria line has 30 links (two directions) and 2,880 eligible observations, carrying 2,975,891 passenger-link trips on 16,228 observed allocations, with up to 10 trains per link per 15 minutes. Its busiest observation is Euston LU → Warren Street at 08:30 (5,877 passengers in 15 minutes). We use $C = 986$ (2009 stock). Table VII summarises the three models.

Model 1. With no minimum frequency the line needs 5,049 allocations (68.9% fewer than observed) at an average utilisation of 59.7% of rated capacity, which is necessarily at most ρ. The observed timetable is comfortably sized for this line: its highest load factor is 66% of rated capacity, so 0 link–intervals exceed the 80% target. The model therefore

TABLE VI
MODEL 1 NETWORK-WIDE: OBSERVED VS. MINIMUM REQUIRED LINK-TRAIN ALLOCATIONS PER DAY, AND PEAK ALLOCATIONS PER 15 MIN ON ANY LINK

Line	Observed	Required	Ratio	Peak obs.	Peak req.
LO Liberty	128	128	100%	1	1
LO Suffragette	1,692	1,672	99%	2	2
LO Weaver	3,612	3,561	99%	2	3
LO Lioness	2,664	2,610	98%	2	1
Elizabeth Line	13,124	7,834	60%	6	7
LO Mildmay	7,040	3,978	56%	4	2
H&C and Circle	14,966	7,810	52%	4	3
LO Windrush	9,070	4,517	50%	5	4
Metropolitan	12,558	5,334	42%	6	3
DLR	18,989	7,801	41%	8	3
District	29,530	11,782	40%	7	5
Bakerloo	11,522	4,389	38%	6	3
Central	26,945	10,159	38%	9	6
Piccadilly	28,435	10,415	37%	7	4
Northern	38,919	13,966	36%	8	8
Waterloo & City	605	213	35%	8	5
Jubilee	20,395	6,906	34%	8	6
Victoria	16,228	5,049	31%	10	8
London Trams	0	0	–	0	0

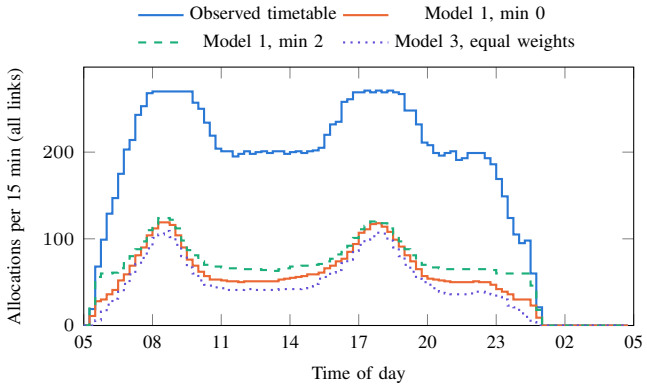


Fig. 9. Victoria line: observed vs. optimised train allocations per 15 minutes, summed over all links.

removes trains rather than adding them, and the peak allocation it needs (8 trains per link per 15 minutes) is close to the 9–10 trains actually run, so the reductions are almost entirely off peak. Requiring at least 2 trains per 15 minutes (a 7.5-minute headway) raises the requirement to 6,046; requiring 4 (3.75 min) gives 9,640 at 31.3% utilisation. Fig. 9 shows that the optimised profiles follow demand closely in the peaks and that the off-peak gap to the timetable is the headway policy.

Model 2. With a budget equal to the observed allocations in each interval, the budget never limits service (0 binding intervals): Model 2 serves all demand with 5,049 allocations, the same total as Model 1. This is expected: the timetable provides more allocations than demand requires in every interval, so the allocation problem reduces to the minimum-frequency problem. With the *include* policy, unmet demand rises to 1,779 passengers. The zero-train records fall in intervals whose observed budget is zero or very small (start and end of the traffic day), so the budget cannot serve them: an apparent shortfall created entirely by the data inconsistency. Reducing the budget (Table VIII, Fig. 10) still serves at least 99.99% of

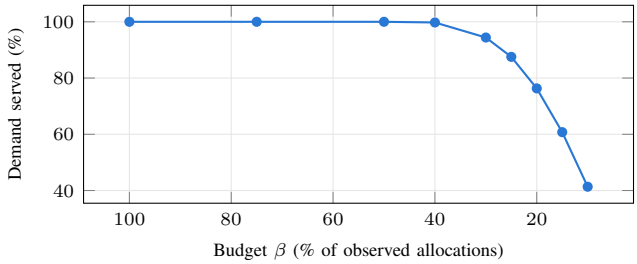


Fig. 10. Model 2: share of demand served as the allocation budget is reduced (Victoria line).

demand down to $\beta = 50\%$, because most observed allocations are not needed to carry volume. Below that, the peak intervals become binding first, then the whole day: at $\beta = 25\%$ the line serves 87.53% of demand with 74 of 96 intervals binding. Within each interval, the optimiser gives the scarce trains to the links where each additional train carries the most passengers.

Model 3. With equal weights the plan uses 4,164 allocations and serves 98.93% of demand. Table IX compares the normalised goal deviations of this plan with those of the observed timetable. Table X shows how the plan responds to the weights; the effort target is $E = 5{,}049$.

The weight experiments show how each goal acts:

- *Efficiency-led* weights give the same plan as equal weights (4,164 allocations). G1 only penalises allocations *above* E, and the equal-weight plan already lies below E (so $z_1 = 0$). Raising w_1 therefore has no further effect.
- The plan lies below E because of G4. Unused target capacity (p_{4i}) is penalised, so in lightly loaded intervals the model accepts a little unmet demand at rated capacity instead of running a train that would be mostly empty.
- *Demand-led* weights restore service to 99.86% served with 4,371 allocations.
- *Utilisation-led* weights cut to 3,811 allocations and 96.56% served.
- *Pattern-led* weights pull the plan towards the timetable. With $w_3 = 5$ the plan stops exactly at E (5,049), where G1 starts to resist. With $w_3 = 20$ it reproduces the observed timetable (16,228 allocations, $z_3 = 0$) at the cost of a large utilisation deviation.

The goal programme thus spans the whole range between the demand-driven minimum and the existing service, and the weights make the trade-off explicit and auditable.

E. Q6: sensitivity analysis

Tables XI and XII report the scenarios for Models 1 and 2 on the Victoria line. For Model 1, a 10% demand increase raises the requirement from 5,049 to 5,383 allocations (+6.6%) and a 20% increase to 5,735 (+13.6%). The increase is less than proportional because many lightly loaded link–intervals need one train whether demand grows or not (integer rounding), while the increase is close to proportional in the peaks. Lowering the utilisation target to 70% needs 5,546 allocations; planning at 100% of rated capacity needs 4,271. With the budget enforced

TABLE VII
VICTORIA LINE: MODEL RESULTS (WHOLE DAY, $\rho = 80\%$)

Case	Allocations	Demand served (%)	Unmet	Utilisation of rated cap. (%)	Links with shortfall	Time (s)
M1, min 0	5,049	100.00	0	59.7	0	0.5
M1, min 2	6,046	100.00	0	49.9	0	0.2
M1, min 4	9,640	100.00	0	31.3	0	0.2
M1, zero-train incl.	5,158	100.00	0	58.5	0	0.3
M2, budget 100%	5,049	100.00	0	59.7	0	0.7
M2, budget 25%	4,023	87.53	370,568	65.6	30	5.0
M3, equal weights	4,164	98.93	31,739	71.6	30	1.3

TABLE VIII
MODEL 2 ON THE VICTORIA LINE: EFFECT OF THE ALLOCATION BUDGET β

β	Allocations	Served (%)	Unmet	Binding int.
100%	5,049	100.00	0	0
75%	5,049	100.00	0	0
50%	5,047	100.00	14	1
40%	4,979	99.76	7,168	11
30%	4,542	94.41	166,136	29
25%	4,023	87.53	370,568	74
20%	3,213	76.31	703,893	79
15%	2,398	60.76	1,166,309	79
10%	1,587	41.35	1,743,001	79

TABLE IX
MODEL 3 (EQUAL WEIGHTS): NORMALISED GOAL DEVIATIONS z_k (0 = GOAL MET)

Goal	w_k	Observed timetable	Optimised plan
G1 efficiency	1	2.214	0.000
G2 demand	1	0.000	0.011
G3 service pattern	1	0.000	0.743
G4 utilisation	1	3.307	0.184

TABLE X
MODEL 3 ON THE VICTORIA LINE: EFFECT OF GOAL WEIGHTS (w_1, w_2, w_3, w_4)

Weights	Alloc.	Served %	z_1	z_2	z_3	z_4
Equal (1,1,1,1)	4,164	98.93	0.000	0.0107	0.743	0.184
Efficiency-led (5,1,1,1)	4,164	98.93	0.000	0.0107	0.743	0.184
Demand-led (1,5,1,1)	4,371	99.86	0.000	0.0014	0.731	0.217
Pattern-led (1,1,5,1)	16,228	100.00	2.214	0.0000	0.000	3.307
Utilisation-led (1,1,1,5)	3,811	96.56	0.000	0.0344	0.765	0.160

TABLE XI
SENSITIVITY ANALYSIS, MODEL 1 (VICTORIA LINE)

Scenario	Alloc.	Served %	Unmet	Util. %	Short. links
A. Baseline	5,049	100.00	0	59.7	0
B. Demand +10%	5,383	100.00	0	61.6	0
C. Demand +20%	5,735	100.00	0	63.1	0
D. Budget -10%	5,049	100.00	0	59.7	0
E. Utilisation 70%	5,546	100.00	0	54.4	0
E. Utilisation 90%	4,596	100.00	0	65.6	0
E. Utilisation 100%	4,271	100.00	0	70.6	0

TABLE XII
SENSITIVITY ANALYSIS, MODEL 2 (VICTORIA LINE, BUDGET CUT 25% IN D)

Scenario	Alloc.	Served %	Unmet	Util. %	Short. links
A. Baseline	5,049	100.00	0	59.7	0
B. Demand +10%	5,383	100.00	0	61.6	0
C. Demand +20%	5,735	100.00	0	63.1	0
D. Budget -25%	5,049	100.00	0	59.7	0
E. Utilisation 70%	5,546	100.00	0	54.4	0
E. Utilisation 90%	4,596	100.00	0	65.6	0
E. Utilisation 100%	4,271	100.00	0	70.6	0

TABLE XIII
SOLVE TIMES (INCLUDING MODEL CONSTRUCTION)

Scope	Model	Obs.	Time (s)	Status
Victoria (30 links)	Model 1	2,771	0.3	Optimal
Victoria (30 links)	Model 2	2,771	0.9	Optimal
Victoria (30 links)	Model 3	2,771	1.4	Optimal
Central + Northern (212 links)	Model 1	19,388	1.8	Optimal
Central + Northern (212 links)	Model 2	19,388	15.6	Optimal
Central + Northern (212 links)	Model 3	19,388	40.3	Optimal
Whole network (1,214 links)	Model 1	110,582	9.9	Optimal

and reduced by 10%, Model 1 remains feasible, because the demand-driven requirement is far below the timetable. For Model 2, demand growth of up to 20% and a 25% budget cut still leave at least 99.99% of demand served on this line. This is consistent with the budget sweep: the Victoria line has enough spare scheduled capacity to absorb both. Lines with high load factors, such as LO Weaver, have far less margin (Table VI).

Infeasibility is reported, not hidden. For example, planning the AM peak (07:00–10:00) at $\rho = 50\%$ with the maximum frequency capped at the observed peak of 10 trains per 15 minutes returns "Infeasible", with the diagnosis: 25 observation(s) need more trains than the maximum frequency (10).

F. Computational performance

Table XIII lists solve times on a laptop (8 threads). Model 1 is solved for the whole network in seconds. Models 2 and 3 are solved to proven optimality for a line in a few seconds; whole-network runs of Models 2 and 3 take minutes and are bounded by a user-set time limit.

VIII. DISCUSSION AND LIMITATIONS

Interpretation. Across the network, the capacity needed to carry the observed volume at an 80% target is well below what is scheduled. Two kinds of mismatch emerge. In the

peaks, specific links exceed the target load, and these are where additional or reallocated trains are justified (Model 1 adds trains there; Model 2 keeps them under budget cuts). Off peak, the timetable carries large spare capacity, which is the cost of short, regular headways. The headway level is a policy choice, not an output of demand data, and it can be expressed through the minimum-frequency constraint. Model 3 makes this trade-off explicit: raising the weight on the existing service pattern keeps the timetable's regularity, while raising efficiency or utilisation weights trims off-peak service.

Limitations.

- Train capacity is an assumption. Capacity bases differ by mode, and a uniform value ignores differences between lines.
- The allocation budget sums link frequencies; it is not an inventory of physical trains.
- Link-level allocation is not a timetable. Neighbouring links may receive different frequencies, and train circulation, terminal turn-backs and crew or rolling-stock schedules are not modelled.
- NUMBAT loads are modelled estimates for a typical autumn day; the cover sheet notes lower confidence on routes that parallel non-TfL services.
- Excluded observations (ambiguous links and, by default, zero-train demand) slightly understate demand on the affected links; the amounts are disclosed.
- The network figure is a schematic topology, not a geographic map.

Future work. Route-level consistency constraints (equal frequency along a service pattern), passenger waiting time as an explicit objective, and day-type comparison (Friday, weekend) are natural extensions.

IX. Conclusion

This project built an end-to-end, data-driven decision-support system for metro train frequency and capacity planning on Transport for London's NUMBAT data. A schema-detecting parser and a strict validation pipeline turn the published workbook into a clean link–interval dataset. The pipeline discloses every exclusion, including link–intervals that report passengers without a scheduled train. Three models answer complementary questions: Model 1 gives the minimum frequency to carry demand, Model 2 the best use of a limited service budget, and Model 3 a weighted compromise between effort, demand, service pattern and utilisation. Scenario analysis shows how these answers respond to demand growth, budget cuts and utilisation targets. The results show that observed peak crowding is localised to identifiable links and intervals, and that most of the remaining gap between demand-driven and timetabled service reflects headway policy. The interactive dashboard makes these trade-offs transparent and reproducible for planners and students alike.

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